Impact of Reconstruction Errors on Connectome Graph Analysis

FlyWire connectome simulation study — BANC dataset

August 11, 2026

Reconstruction-error simulation (missed vs. false synapses) on a real *Drosophila* connectome; 10 error rates × 5 seeded trials per model; 49 graph metrics compared against the original network.

1 Question

Connectome reconstruction from electron-microscopy data is never perfect: segmentation and synapse-detection algorithms make mistakes. This raises a practical question for every downstream analysis:

> ***How much do reconstruction errors distort the results of connectome graph analysis, and do different error types distort it differently?***

Concretely, we simulate two canonical error types on the BANC connectome and answer three sub-questions:

1. **Direct effect:** how do total synapses and connections change with the error rate?
2. **Structural effect:** are global properties (connectivity, reciprocity, hub ranking) preserved, or do they degrade?
3. **Comparative effect:** which error type — missed synapses (false negatives) or false synapses (false positives) — is more disruptive at the same nominal rate?

2 Method

2.1 Data

The BANC (*Brain And Nerve Cord*) connectome of *Drosophila melanogaster*: 158,262 neurons, 3,990,039 directed connections, 23,556,214 synapses. This is the ground-truth network used as the baseline for all comparisons.

2.2 Error models

Two stochastic perturbation models are applied to the original graph (seeded, reproducible; 5 trials per configuration):

Missed synapses (false negatives). Every synapse survives independently with probability $1 - r$ (binomial sampling). An edge disappears only when *all* of its synapses are dropped; otherwise its weight shrinks. Synapse loss is therefore approximately linear in r, while edge loss is slow.

False synapses (false positives). $k = r \times$ (original edge count) artificial connections are injected between neuron pairs that had no connection, with weights drawn from the weak-edge distribution (≤ 5 synapses). Edge and synapse counts grow approximately linearly with r.

2.3 Experimental design

- Error rates: 0, 0.25, 0.5, 0.75, 1, 2, 5, 10, 15, 20%.
- 5 seeded trials per rate per model (seeds recorded in metadata).
- 49 metrics per trial, grouped into 6 analysis families: structure, degree distributions, connected components, reciprocity, assortativity, and PageRank.
- Each metric is converted to a *preservation score* Preservation $= \dfrac{\min(B, P)}{\max(B, P)} \times 100\%$ between baseline B and perturbed value P. 14 of the 49 metrics have a well-defined baseline and enter the preservation analysis; the remaining 35 (KS/Wasserstein distances, PageRank score rows) have baseline 0 and are reported as distances.

3 Results

3.1 R1: Direct effects are exactly proportional to the error rate

Figure 1 shows the cleanest signal of the study: the two models change synapse and edge counts *linearly* in the error rate. Missed synapses remove -20.0% of synapses at 20% rate, but only -3.2% of connections — the coarse wiring survives because an edge is only lost when every one of its synapses is missed. False synapses do the opposite: connections grow by +20.0%, synapses by +10.0%.

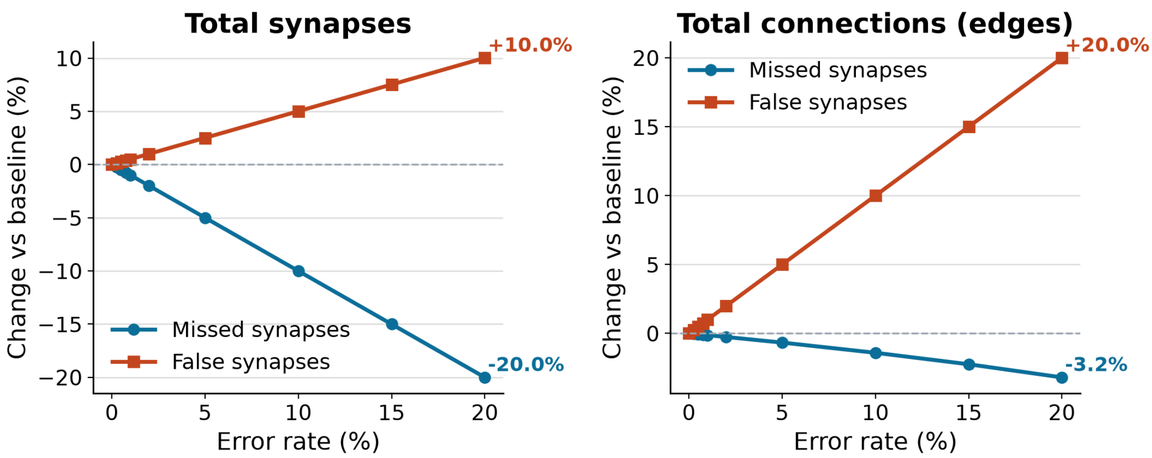

Figure 1: Change vs. baseline against the error rate. Both error types act proportionally and predictably: missed synapses deflate the synapse pool, false synapses inflate the connection count. Axes are identical across panels and lines are annotated at the 20% endpoint.

Table 1: Snapshot at 20% error — baseline vs. the two perturbed networks (mean of 5 trials).

Metric	Baseline	Missed @20%	False @20%	Missed change	False change
Total synapses	23,556,214	18,843,554	25,917,579	-20.0%	+10.0%
Connections (edges)	3,990,039	3,862,712	4,788,047	-3.2%	+20.0%
Reciprocity	0.1299	0.1292	0.1391	-0.6%	+7.1%
Giant SCC size	139,405	139,342	139,957	-0.05%	+0.40%

3.2 R2: The coarse network architecture is highly robust

Even at 20% error the global wiring is nearly untouched (Figure 2): the giant strongly connected component keeps 99.95% of its vertices under missed synapses and grows by 0.40% under false synapses, and reciprocity moves by only -0.6% / +7.1%. PageRank correlation with the baseline stays at 0.9975 (missed) and 0.9833 (false), and the overlap of the top-ranked neurons drops only to 0.970 / 0.892 at 20% (Figure 3).

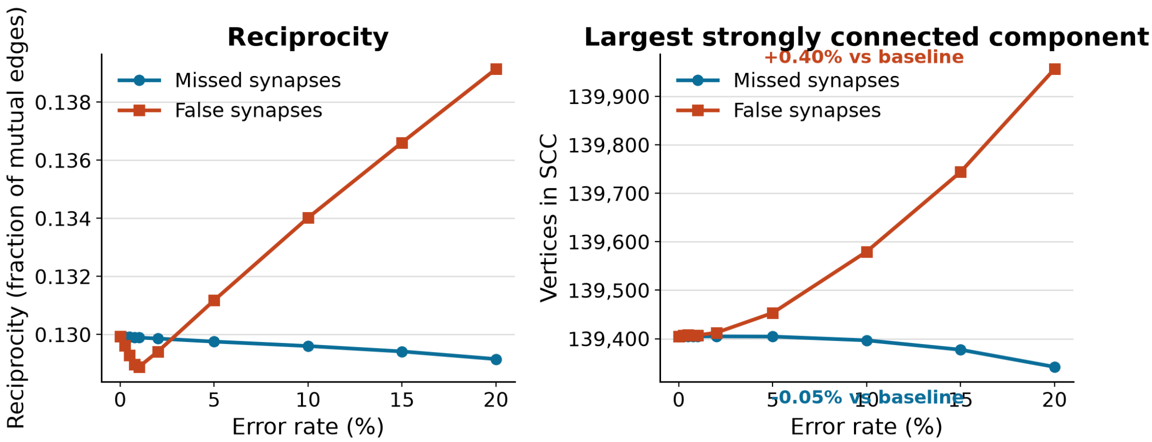

Figure 2: Network organization metrics (raw values). The giant SCC is essentially unchanged; reciprocity drifts only slightly. Y-axes use plain thousands-separated ticks and annotations give the change at 20%.

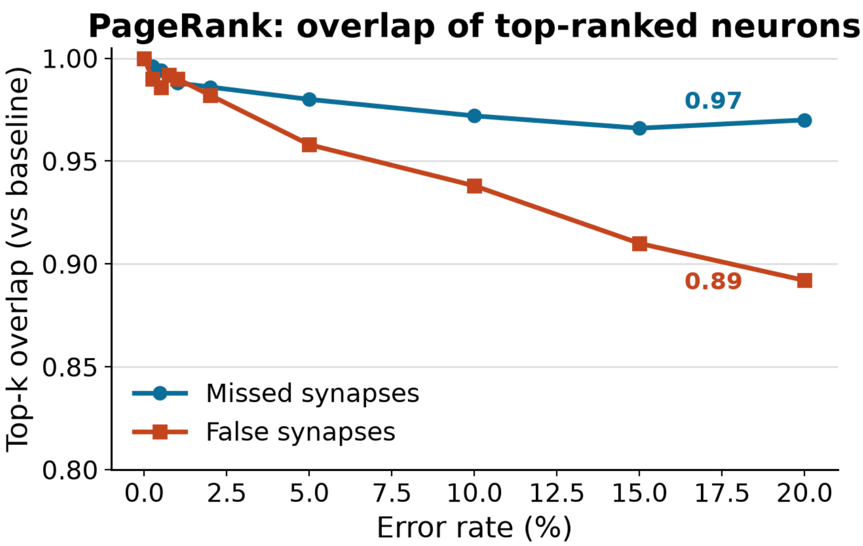

Figure 3: PageRank fidelity: overlap between the top-ranked neurons of the perturbed and baseline networks. The ranking survives both models, with false synapses eroding it more (0.892 vs. 0.970 at 20%).

3.3 R3: False synapses distort degree distributions far more

The most pronounced difference between the models appears in the degree-distribution distances (Figure 4). Because the figure spans more than an order of magnitude, the y-axis is logarithmic so both curves remain readable; the values at 20% are annotated. At equal nominal rates, false synapses inflate the in-degree KS distance by 14.5× and the Wasserstein distance by 6.3× compared with missed synapses.

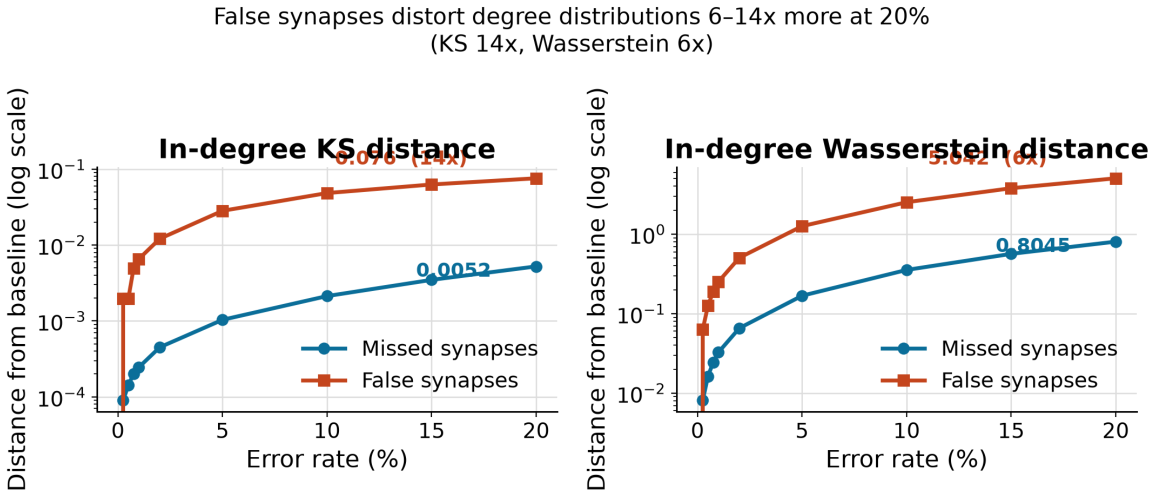

Figure 4: Degree-distribution distortion (log scale). Log axis chosen so the two curves — which differ by an order of magnitude — are both visible; ratios at the 20% endpoint are marked.

3.4 R4: Sensitivity ranking — the synaptic layer degrades first

When every metric is converted to its minimum preservation across all rates (Figure 5, Table 2), a clear ordering emerges: weight statistics and synapse counts are the first to erode under missed synapses (weight variance down to 68.9%), while edge count and density degrade most under false synapses (83.3%). Connectivity, reciprocity, assortativity and PageRank stay above 93% under both models.

Table 2: Minimum preservation across all error rates, most sensitive metrics first.

Metric	Missed	False
Weight median	75.00%	75.00%
Density	96.81%	83.33%
Edge count	96.81%	83.33%
Weight variance	68.95%	84.97%
Total synapses	79.99%	90.89%
Weight mean	82.63%	91.69%
Weight std	83.03%	92.18%
Reciprocity	99.40%	93.38%
Assortativity	99.67%	98.51%
SCC max size	99.95%	99.61%
Weight max	92.11%	100.00%
WCC max size	99.99%	100.00%

Grouping the preservation scores by biological category at the 20% rate (Figure 6, Table 3) shows the same story at the level of whole categories: *Synaptic Properties* is the most affected

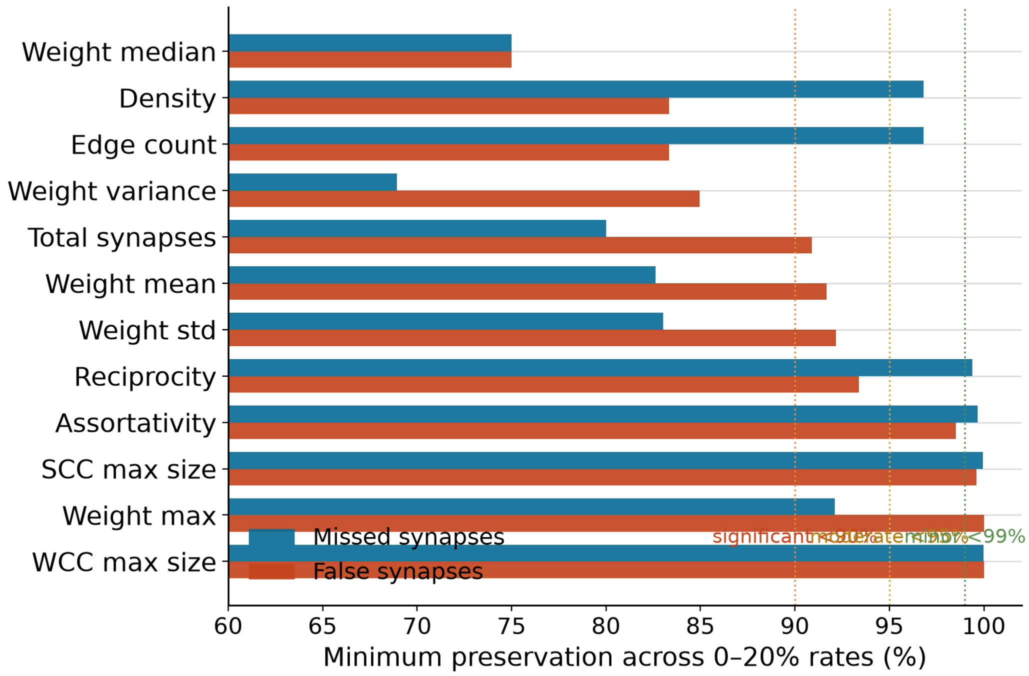

Figure 5: Minimum preservation of each metric across 0–20% rates (grouped by model). Threshold guides: significant <90%, moderate <95%, minor <99%.

category under missed synapses (83.1%), *Structural Topology* under false synapses (88.9%), while *Connectivity* stays above 99.8% in both models.

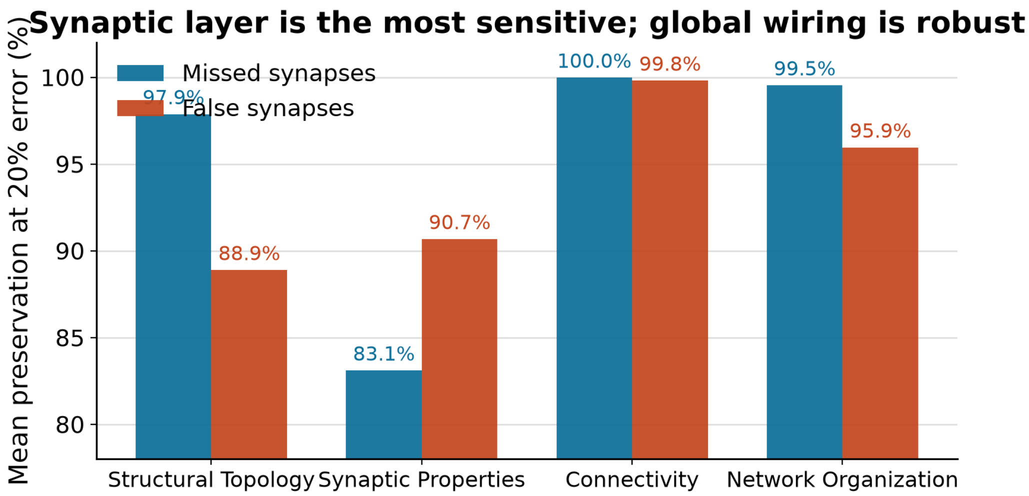

Figure 6: Mean preservation by biological category at 20% error. Value labels are printed on every bar.

Table 3: Category-wise preservation at 20% error (unweighted mean of the member metrics).

Category	Member metrics	Missed	False
Structural Topology	3	97.9%	88.9%
Synaptic Properties	7	83.1%	90.7%
Connectivity	2	100.0%	99.8%
Network Organization	2	99.5%	95.9%

3.5 R5: Results are reproducible across trials

Five seeded trials per configuration give very tight spread (Table 4): the relative standard deviation of total synapses at 20% is 0.009% (missed) and 0.005% (false), and false-synapse trials are essentially deterministic (edge-count $\sigma = 0$). Primary trends are monotone across all ten rates.

Table 4: Relative spread (σ/mean) across the 5 trials at 20% error.

Metric	Missed	False
Total synapses	±0.009%	±0.005%
Edge count	±0.005%	±0.000%
Giant SCC	±0.010%	±0.006%

4 Takeaway

1. **Error effects are direct and predictable.** Missed synapses remove synapses in proportion to the rate; false synapses add connections in proportion to the rate. Both models are linear in the perturbed quantity they control.
2. **The coarse connectome architecture is robust.** Node count, giant connected component, reciprocity and PageRank correlation survive to ≥93% (mostly ≥99%) even at 20% error — the coarse wiring plan is barely disturbed.
3. **The synaptic layer degrades first.** Weight statistics and synapse counts are the most sensitive metrics, dropping to 69–80% preservation under missed synapses at the highest rates.
4. **False synapses are the more disruptive model.** At the same nominal rate they distort degree distributions 6–14× more, erode hub ranking more (top-k 0.892 vs. 0.970), and raise reciprocity by +7.1%, because the injected weak edges (average 2.96 synapses vs. baseline mean 5.90) perturb the graph's fine structure without creating new hubs.
5. **Error signatures are separable.** Missed synapses *deflate* synaptic statistics while leaving topology intact; false synapses *inflate* edge counts and distort degree distributions. An observer can distinguish the two error types from the metric profile alone.

Takeaway: Reconstruction errors on the connectome are a *quantitative, local* problem, not a *qualitative, global* one: synaptic-level statistics and degree distributions respond first and strongly, while the global architecture (connectivity, ranking, reciprocity) stays intact up to 20% error. When interpreting connectome analyses, report synaptic-layer metrics with error-rate caveats — and expect false-positive (spurious-edge) errors to be the more damaging of the two.

Reproducibility

All figures and tables are derived from the aggregated experiment outputs (nothing is hand-typed):

```
# 1. Regenerate the 8 figures from the raw CSVs
python make_figures.py

# 2. Regenerate every number and table used in this report
python make_report_data.py # -> report_data.tex, table1..4.tex

# 3. Compile (twice for cross-references)
pdflatex report.tex
```

Data sources:

- `results/BANC/missed_synapses/.../trend_analysis/combined_results.csv`
- `results/BANC/false_synapses/.../trend_analysis/combined_results.csv`

The preservation scores for the weight and assortativity metrics are recomputed with the framework's current formula (`presentation/preservation_config.py`); see `correction.md` for the provenance of this correction.